LESSON 9.4

Apply Two-Variable Relationships to Solve Problems

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can write an equation to represent the relationship between two variable quantities and use it to solve a real problem.

WHY THIS MATTERS An automobile plant's assembly line runs in 8-hour shifts and produces 2.5 cars per hour. An order arrives: 1,000 cars in 3 weeks. Should the manufacturer accept? Later, the student council plans a fundraiser car wash — last year they washed 34 cars and raised $408, and this year the goal is $500.

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
solution *Spanish: solución*	A value of the variable that makes an equation true.	the number you solve for
substitute *Spanish: sustituir*	To replace a variable with a number so you can compute.	put 8 in for x
at least *Spanish: al menos*	That amount or more — the smallest amount that still works.	“At least $500” means $500 counts, and so does anything more.
predict *Spanish: predecir*	To make a reasonable guess about a future amount using information you already have.	We measured up to 5 minutes → 6,000 feet. Reach PAST what was measured: 6 minutes should be about 7,200.
justify *Spanish: justificar*	To explain WHY an answer or recommendation makes sense, using the mathematics.	“35 cars at $14.30 raises $500.50, which clears the $500 goal.”
Equation *Spanish: Ecuación*	A math sentence with an equals sign showing two amounts are the same.	c = 4n

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

A bakery makes 24 rolls each hour: r = 24h. Is 5 hours enough for 150 rolls?

1. Substitute h = 5: r = 24 × 5 = 120 rolls.
2. Compare: 120 < 150.

Answer: No, 5 hours makes only 120 rolls

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Last year the council washed 34 cars and raised $408. What did they charge per car? Divide: 408 ÷ 34 = $12.

1. _____
2. _____
3. _____
4. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- I let _____ stand for _____ and _____ stand for _____ .
- My equation is _____ , so I substitute _____ for _____ .
- The solution is _____ , which means _____ in this situation.
- This answer is reasonable because _____ .

Word bank equation · substitute · solution · rate · variable · multiply · per · total · reasonable · check · input · output

WATCH OUT!

A common mistake

Adding the rate instead of multiplying

LESSON 9.4

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Version A · Supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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WHY THIS MATTERS An automobile plant's assembly line runs in 8-hour shifts and produces 2.5 cars per hour. An order arrives: 1,000 cars in 3 weeks. Should the manufacturer accept? Later, the student council plans a fundraiser car wash — last year they washed 34 cars and raised $408, and this year the goal is $500.

1 **Circle the letter of the best answer. Show how you know.**

The assembly line produces 2.5 cars per hour. How many cars does it produce in one 8-hour shift?

(A) 10.5 cars
(B) 16 cars
(C) 20 cars
(D) 25 cars

HINT How many cars come off the line in ONE hour?

PER 1	

1. **Label** *Write what each column is.*
2. **Find per 1** *Divide to get one.*
3. **Scale** *Multiply up to what is asked.*
4. **Check** *Does the size make sense?*

2 **Circle the letter of the best answer. Show how you know.**

Let h = hours of operation and c = cars produced. Which equation represents the assembly line?

(A) c = 2.5h
(B) c = 2.5 + h
(C) h = 2.5c
(D) c = h ÷ 2.5

HINT Say it in words: the cars produced equal 2.5 ___ the hours.

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

Using the table (1 hour, 2.5 cars; 8 hours, 20 cars; 16 hours, 40 cars), how many cars are produced in 24 hours?

(A) 26.5 cars

(B) 40 cars

(C) 50 cars

(D) 60 cars

HINT How many 8-hour shifts fit in 24 hours?

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Last year the student council washed 34 cars and raised $408. How much did they charge per car?

(A) $10

(B) $12

(C) $34

(D) $374

HINT The total is the price per car times the number of cars.

PER 1	

1. **Label** *Write what each column is.*
2. **Find per 1** *Divide to get one.*
3. **Scale** *Multiply up to what is asked.*
4. **Check** *Does the size make sense?*

5 **Circle the letter of the best answer. Show how you know.**

At $12 per car, how much does the student council raise by washing 20 cars?

(A) $32
(B) $120
(C) $200
(D) $240

HINT Each car adds $12.

PER 1	

1. **Label** *Write what each column is.*
2. **Find per 1** *Divide to get one.*
3. **Scale** *Multiply up to what is asked.*
4. **Check** *Does the size make sense?*

6 **Circle the letter of the best answer. Show how you know.**

The price is set at $12 per car. Which is the DEPENDENT variable at the car wash?

(A) The number of cars washed
(B) The total amount of money raised
(C) The price of $12 per car
(D) The number of students on the council

HINT Which quantity responds to the other?

SHOW YOUR WORK

EXPLAIN YOUR THINKING

One rate is cars per hour and the other is dollars per car. Ask your partner to predict one value from each, then check by substituting — which prediction could you make without a calculator, and why?

SENTENCE STARTER Substituting ___ gives ___, so I predict ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 9.4

Apply Two-Variable Relationships to Solve Problems

Version B · Core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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1 **Circle the letter of the best answer. Show how you know.**

The order requires 1,000 cars. Using 2.5h = 1,000, how many hours will the plant need?

(A) 40 hours
(B) 400 hours
(C) 997.5 hours
(D) 2,500 hours

My work — one step per line	What I did to both sides

2 **Circle the letter of the best answer. Show how you know.**

The plant needs 400 hours to fill the order. Running 24 hours a day, about how many days is that?

(A) 4 days
(B) About 17 days
(C) 50 days
(D) 400 days

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

The goal is $500 at $12 per car. Using 12n = 500, at least how many cars must be washed?

(A) 40 cars
(B) 41 cars
(C) 42 cars
(D) 50 cars

PER 1

4 **Circle the letter of the best answer. Show how you know.**

Can the student council wash 41 cars at $12 each and meet the $500 goal?

(A) No — 41 cars raise $492, which is less than $500
(B) Yes — 41 cars raise $502
(C) Yes — 41 rounds up to 42, so it counts
(D) No — 41 cars raise $412

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

The council expects 35 cars and wants to raise at least $500. What is the minimum price per car, in whole cents, that reaches the goal? Use 35p ≥ 500.

(A) $12.00
(B) $14.00
(C) $14.29
(D) $465.00

PER 1

6 **Circle the letter of the best answer. Show how you know.**

How could the council CHECK that charging $14.30 for 35 cars meets the goal?

(A) Divide 35 by 14.30 and confirm it is at least $500

(B) Multiply 35 × 14.30 = 500.50 and confirm it is at least $500

(C) Add 35 + 14.30 and confirm it is at least $500

(D) No check is possible after solving

SHOW YOUR WORK

EXPLAIN YOUR THINKING

One rate is cars per hour and the other is dollars per car. Ask your partner to predict one value from each, then check by substituting — which prediction could you make without a calculator, and why?

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Challenge · Extension

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- ☐ Exceeds
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PART 1 Understand the idea

models and meaning

1 **Write the letter of the matching item on each line.**

Connect the four representations of the assembly line: scenario, equation, table, and graph.

_____ Scenario: 2.5 cars every hour

_____ Table row: h = 8, c = 20

_____ Graph point (400, 1000)

_____ Equation c = 20s

(A) Equation: c = 2.5h

(B) Graph point (8, 20)

(C) The 1,000-car order takes 400 hours

(D) Cars counted by 8-hour shifts

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

Suppose the variable s counts 8-hour SHIFTS instead of hours. Which equation gives the cars produced?

(A) c = 2.5s

(B) c = 8s

(C) c = 10.5s

(D) c = 20s

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

If the plant runs 24 hours a day, 7 days a week for the full 3 weeks, how many hours is that — and can it fill the 1,000-car order?

(A) 72 hours — no, that is less than 400 hours

(B) 168 hours — no, that is less than 400 hours

(C) 504 hours — yes, because 504 hours is more than the 400 hours needed

(D) 504 hours — no, because 504 is less than 1,000

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

The solution to one problem was 41.7 cars. Would producing or washing a PARTIAL car make sense?

(A) No — cars come in whole numbers, so round up to the next whole car

(B) Yes — 0.7 of a car is a real amount you can wash

(C) No — so round down and accept a smaller total

(D) Yes — the equation is wrong if it gives a decimal

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

Would you recommend the student council charge $12 or $14.30 per car? Which reasoning is strongest?

(A) Either can work: $12 needs at least 42 cars washed, while $14.30 needs only about 35 — so the choice depends on how many customers they expect

(B) $12 is always better because a smaller number is always better

(C) $14.30 is always better because more money per car is always better

(D) Neither price can reach the $500 goal

SHOW YOUR WORK

EXPLAIN YOUR THINKING

One rate is cars per hour and the other is dollars per car. Ask your partner to predict one value from each, then check by substituting — which prediction could you make without a calculator, and why?

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